CURRICULUM VITAE

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RESEARCH INTERESTS

I am primarily interested in the dimension theory and geometry of fractal sets and measures. I often work with dynamically defined fractals such as attractors of iterated function systems. I am interested in finding links between fractal geometry and other areas of mathematics including: real and complex analysis, hyperbolic geometry, Fourier analysis, geometric measure theory, ergodic theory, and probability theory.

EMPLOYMENT

- Aug 2021 – present: Professor of Mathematics and Statistics, University of St Andrews.
- Aug 2018 – July 2021: Reader in Mathematics and Statistics, University of St Andrews.
- Aug 2016 – July 2018: Lecturer in Mathematics and Statistics, University of St Andrews.
- Sept 2014 – July 2016: Lecturer in Pure Mathematics, University of Manchester.
- July 2013 – Aug 2014: Research Fellow, University of Warwick.
  - Funded by Mark Pollicott's EPSRC grant EP/J013560/1 *Thermodynamic formalism and flows on moduli spaces.*

EDUCATION

- Sept 2009 – Jun 2013: PhD in Pure Mathematics, University of St Andrews (EPSRC funded).
  - Thesis: *Dimension theory and fractal constructions based on self-affine carpets.*
- Sept 2005 – May 2009: MMath (with Hons), First Class, University of St Andrews.
  - Honours average: 19.0/20

SELECTED AWARDS AND PRIZES

Research and scholarship

- 2026: *Wallenberg Visiting Professor*, Institut Mittag-Leffler.
- 2024: *Fellow of the Royal Society of Edinburgh* (FRSE).
- 2023: *Sir Edmund Whittaker Memorial Prize*, Edinburgh Mathematical Society.
- 2022: *Sir Thomas Makdougall Brisbane Medal*, Royal Society of Edinburgh early career medal in physical, engineering & informatic sciences (including mathematics).
- 2018: *Young Academy of Scotland.*

Teaching awards

- 2023: *Outstanding Teacher (Science & Medicine)* - winner, Students' Association Teaching Awards.
- 2015: *Best Lecturer within the Faculty of Engineering and Physical Sciences* (Uni. Manchester).
- 2013: *Best Postgraduate Tutor* - St Andrews Students' Association Teaching Awards.

Undergraduate prizes (for posterity!)

- 2009: *Class Medal* for best student in 4th/5th level MMath mathematics.
- 2009: *Arthur Hinton Read Memorial Prize* for best student in senior honours pure mathematics.
- 2009: *IMA prize* for best student in senior honours.
- 2008: *Class Medal* for best student in 3rd level MMath mathematics.
- 2008: *Alexander Stewart Prize* for best student in junior honours pure mathematics.

FINANCIAL SUPPORT

Total grant income of £2,739,646 since 2015 including:

- 2025–30: EPSRC Open Fellowship — £1,487,240
  - *A new theory of dimension interpolation*, EP/Z533440/1
- 2024–28: Leverhulme Trust Research Project Grant (PI) — £377,327
  - *A refined Fourier analysis*, RPG-2023-281
- 2024–25: EPSRC Standard Grant (PI) — £50,597
  - *Fourier analytic techniques in finite fields*, EP/Y029550/1
- 2021–22: RSE Sabbatical Research Grant — £62,916
  - *Fractal dimension in nonlinear nonconformal settings*, award number: 70249
- 2020: ICMS Workshop grant — £29,000
- 2019–23: Leverhulme Trust Research Project Grant (PI) — £324,135
  - *New perspectives in the dimension theory of fractals*, RPG-2019-034
- 2018–21: EPSRC Standard Grant (PI) — £336,123
  - *Fourier analytic techniques in analysis and geometry*, EP/R015104/1
- 2016–18: Leverhulme Trust Research Fellowship — £48,003
  - *Fractal geometry and dimension theory*, RF-2016-500
- Several smaller grants, including from: LMS, RSE, EMS, Scottish Crucible, Glasgow Mathematical Journal Trust, Oberwolfach Research Institute for Mathematics, and the Institut Mittag-Leffler.

RESEARCH ACTIVITY

Complete lists of publications and talks are provided at the end of this CV.

- Written 92 research articles, 9 survey articles, and a book entitled *Assouad Dimension and Fractal Geometry* - CUP, *Tracts in Mathematics Series*, **222**, (2020). To date, 85 research articles accepted for publication by internationally leading peer reviewed journals including:

- Advances in Mathematics (×5)
- Analysis & PDE
- Bulletin of the AMS
- Bulletin of the LMS (×3)
- Ergodic Theory and Dynamical Systems (×7)
- Geometriae Dedicata
- Indiana University Mathematics Journal (×7)
- International Mathematics Research Notices (×2)
- Israel Journal of Mathematics (×2)
- Journal of the LMS (×2)
- Mathematische Annalen
- Mathematische Zeitschrift (×5)
- Nonlinearity (×6)
- Proceedings of the AMS (×10)
- Proceedings of the LMS
- Transactions of the AMS (×4)

- Over 2100 citations and an h-index of 26 (Google Scholar).
- Supervised 12 PhD students, 5 postdoctoral researchers, and written papers with over 50 different collaborators.
- Invited longer term research visits include: Mittag-Leffler Institute (2017 and 2026) and ICERM Brown (2016).
- Given over 80 (external) research talks ∘ regularly invited to speak at large international conferences, including the opening keynote lecture at *Fractal Geometry and Stochastics VI* in 2018 ∘ numerous seminars in the UK and abroad, including at: Warwick, Edinburgh, Imperial, Hebrew U. Jerusalem, Chinese U. Hong Kong, Waterloo (Canada), Stony Brook (USA), Vienna, CNRS Paris, East China Normal University, Helsinki, Oulu, and Bremen.

SUPERVISION AND MENTORSHIP

Postdoctoral researchers supervised

- Roope Anttila, 2025– : funded by *EPSRC* Open Fellowship EP/Z533440/1.
- Firdavs Rakhmonov, 2024– : funded by *Leverhulme Trust* Research Project Grant RPG-2023-281.
- Natalia Jurga, 2020–22 : funded by *EPSRC* Standard Grant EP/R015104/1.
  - Natalia took up a *Leverhulme Trust Early Career Fellowship* in May 2022 and is now a Lecturer in St Andrews.
- István Kolossváry, 2019–23: funded by *Leverhulme Trust* Research Project Grant RPG-2019-034.
  - Istvan took up a *Marie Curie Fellowship* in Budapest in January 2024.
- Jonathan Hickman, 2018–19 : funded by *EPSRC* Standard Grant EP/R015104/1.
  - Jonathan took up a Lectureship at the University of Edinburgh in August 2019.

PhD students supervised

- Tianyi Feng, 2024– : funded by *Leverhulme Trust* Research Project Grant RPG-2023-281.
- Yunlong Xu, 2024– : funded by the *Chinese Scholarship Council*.
- Lijian Yang, 2024– : *Handsel scholarship*.
- Ana Emilia de Orellana, 2023– : *Handsel scholarship*.
- Alex Rutar, 2020–24 : funded by the University of St Andrews and *NSERC*.
  - Thesis: *Assouad-type dimensions and the local geometry of fractal sets*.
  - Alex took up a postdoctoral research position at Uni. Jyväskylä in September 2024.
- Amlan Banaji, 2019–23 : funded by *Leverhulme Trust* Research Project Grant RPG-2019-034.
  - Thesis: *Interpolating between Hausdorff and box dimension*.
  - Amlan took up a Research Associate position at Uni. Loughborough in March 2023.
- Liam Stuart, 2019–22 : funded by the University of St Andrews.
  - Thesis: *Limit sets, Julia sets and Sullivan's dictionary: a dimension theoretic analysis*.
  - Liam took up a Research Fellowship in St Andrews in October 2022.
- Haipeng Chen, 2019–20 : visit funded by the *Chinese Scholarship Council*.
  - Thesis: *Lower Assouad type dimensions of fractal sets* (South China University of Technology)
  - Haipeng took up a Tenure-track Assistant Professorship at the Shenzhen Technology Uni. in 2021.
- Stuart Burrell, 2017–21 : funded by the *Carnegie Trust*.
  - Thesis: *Coincidence and disparity of dimensions in four fractal families*.
  - Stuart took up an LMS Early Career Postdoctoral Fellowship at Uni. St Andrews in January 2021
- Lawrence Lee, 2017–21 : funded by the *EPSRC*.
  - Thesis: *Multifractal measures - from self-affine to nonlinear*.
  - Lawrence took up an EPSRC Doctoral Prize Fellowship at the Uni. Manchester in April 2021.
- Douglas Howroyd, 2016–20 : funded by the *EPSRC*.
  - Thesis: *On the regularity dimensions of measures*.
  - Douglas took up a position as Software Engineer at the Home Office in 2020.
- Han Yu, 2015–19 : funded by the University of St Andrews.
  - Thesis: *Assouad type dimensions and dimension spectra*.
  - Han took up a three year postdoctoral position at the University of Cambridge in September 2019, as well as a Research Fellowship in Corpus Christi College.

Other researchers supervised

- Andrew Mitchell, 2018–19 : funded by a *Scottish Crucible* research grant.
  - Andrew took up a funded PhD position at the University of Birmingham in September 2019.

SELECTED ACADEMIC RESPONSIBILITIES

Membership of Editorial Boards

- 2022– : Editor-in-Chief, *Proceedings of the Royal Society of Edinburgh (Section A, Mathematics)*
- 2022– : *Journal of Fractal Geometry*
- 2020–24: *Royal Society Open Science*
- 2017–23: *Newsletter of the London Mathematical Society*
  - March 2021– : author of the *Mathematics News Flash* section of the bi-monthly Newsletter.
- 2015– : *Proceedings of the Royal Society of Edinburgh (Section A, Mathematics)*

External appointments

- 2024–28: External examiner in Analysis, University of Warwick.
- December 2023: EPSRC Mathematical Sciences Fellowship Interviews Panel.
- 2023: EPSRC CDT reviewer.
- 2022: Swedish Research Council - Mathematics Panel.
- 2022: YAS recruitment - Panel 6: Informatics, Mathematics, Computing and Statistics.
- January 2022: Pure mathematics hiring committee (external member), University of Glasgow.
- 2020–22: REF 2021 Sub-panel 10: Mathematical Sciences.
- January 2020: EPSRC Mathematical Sciences Fellowship Interviews Panel.
- 2020– : Full Member on the *Scottish Mathematical Council.*
- 2018– : EMS representative, London Mathematical Society *International Affairs Committee.*
- 2018–23: *Young Academy of Scotland.*
- 2017–19: *EPSRC Mathematical Sciences Early Career Forum.*
- 2017: *Scottish Crucible.*

Society memberships

- 2010– : *London Mathematical Society.*
- 2009– : *Edinburgh Mathematical Society.*
- 2009–10: *Institute for Mathematics and its Applications.*

Refereeing and reviewing

- Refereed over 90 research articles for over 43 different research journals, including: ∘ *Analysis & PDE* ∘ *Annals of Mathematics* ∘ *Duke Mathematical Journal* ∘ *IMRN* ∘ *Inventiones Mathematicae* ∘ *Journal of the European Mathematical Society (JEMS)* ∘ *Probability Theory and Related Fields (PTRF)* ∘ *Proceedings of the London Mathematical Society* ∘ *Transactions of the American Mathematical Society.*
- Refereed several research proposals, including for: ∘ ***Austrian*** *Science Fund (FWF)* ∘ *Banff International Research Centre (BIRS),* ***Canadian*** ∘ *EPSRC,* ***UK*** ∘ ***European*** *Women in Mathematics/ European Mathematical Society* ∘ ***European*** *Research Council (ERC)* ∘ *The Fund for Scientific Research (FNRS) –* ***Belgium*** ∘ ***German*** *Academic Exchange Service (DAAD)* ∘ ***Israel*** *Science Foundation (ISF)* ∘ *Leverhulme Trust,* ***UK*** ∘ *Marsden Fund - Royal Society of* ***New Zealand*** ∘ *National Research, Development and Innovation Office of* ***Hungary*** ∘ *National Science Centre* ***Poland*** ∘ *Natural Sciences and Engineering Research Council of* ***Canada*** ∘ ***South Africa****'s National Research Foundation* ∘ ***Swiss*** *National Science Foundation*
- Refereed several book proposals, including for: ∘ *CRC Press (Taylor and Francis)* ∘ *Oxford University Press* ∘ *Springer Undergraduate Mathematics Series.*
- 2012– : regular reviewer for the *American Mathematical Society Mathematical Reviews* (60 reviews on MathSciNet to date).

PhD examination

• 2025: Boyuan Zhao (Uni. St Andrews) - *Exponential mixing and almost sure limit theorems in dynamical systems.*
• 2024: Raquel Couto (Uni. Porto, Portugal & Uni. St Andrews) - *Statistical properties and rare events for chaotic dynamical systems.*
• 2024: Tom Rush (Uni. Bristol) - *Thermodynamic formalism for iterated function systems.*
• 2019: Alexandros Margaris (Uni. Warwick) - *Dimensions, Embeddings and Iterated Function Systems.*
• 2018: Demi Allen (Uni. York) - *Mass Transference Principles and Applications in Diophantine Approximation.*
• 2016: Vuksan Mijovic (University of St Andrews) - *Multifractal Zeta Functions.*
• 2015: Eino Rossi (Uni. of Jyväskylä, Finland) - *Local structure of fractal sets: tangents and dimension.*
• 2015: Anthony Chiu (Uni. Manchester) - *Iterated function systems that contract on average.*

Selected departmental and University roles

• 2024 : Director of Research.
• 2022– : member of School Postgraduate Research Committee.
• 2020– : member of School Research Committee (chair 2024).
• 2020–24 : Honours Advisor of Studies in the School of Mathematics and Statistics.
• 2019– : Head of Analysis Research Group.
• 2018–24 : member of University Academic Advising Best Practise Group.
• 2018–23 : Outreach Officer in the School of Mathematics and Statistics.
• 2016–20 : Sub-honours Advisor of Studies in the School of Mathematics and Statistics.
• 2016– : member of various shortlisting and interview panels at Manchester and St Andrews.

TEACHING

Undergraduate modules lectured since joining St Andrews in 2016:
◦ MT1002: Mathematics - 2017
◦ MT2502: Analysis - 2019, 2020, 2023, 2024
◦ MT3502: Real Analysis - 2017, 2018
◦ MT5830/MT5870: Hyperbolic Geometry - 2017, 2019, 2021, 2023, 2025
◦ MT5877: Ergodic Theory and Dynamical Systems - 2024

I wrote the Hyperbolic Geometry course and a full set of illustrated lecture notes is available upon request. I have always received excellent student feedback (both unofficial and official). The formal feedback I have received is summarised below, where the numbers are average scores given by students on scale of 1-5 with 1 the best. There are four categories directly relating to the lecturer, two of which changed from 2019: ◦ Lecture material was well organised (organisation) ◦ Lecturer was good at explaining things/Lecture material was well explained (explanation) ◦ Teaching style was engaging/Lecture material was well presented (presentation) ◦ I was able to contact the lecturer if I needed to (availability).

	organisation	presentation	explanation	availability	class size
MT5870 (Spring 2025)	1.0	1.33	1.22	1.22	32
MT2502 (Fall 2024)	1.17	1.04	1.22	1.26	134
MT5877 (Spring 2024)	1.57	1.14	1.14	1.67	15
MT2502 (Fall 2023)	1.14	1.19	1.19	1.28	129
MT5870 (Spring 2023)	1.14	1.21	1.07	1.14	35
MT5870 (Spring 2021)	1.36	1.18	1.36	1.0	27
MT2502 (Fall 2020)	1.32	1.25	1.33	1.42	132
MT2502 (Fall 2019)	1.18	1.16	1.13	1.27	142
MT5830 (Spring 2019)	1.29	1.18	1.18	1.24	29
MT3502 (Fall 2018)	1.27	1.05	1.11	1.26	82
MT3502 (Fall 2017)	1.13	1.03	1.1	1.33	66
MT1002 (Fall 2017)	1.28	1.18	1.26	1.42	213
MT5830 (Spring 2017)	1.4	1.07	1.13	1.13	28

Undergraduate modules taught while at Uni. Manchester:
◦ 0C1/1C1: Mathematics for Optometry and Foundation Year - 2014, 2015
◦ MATH10101: Sets Numbers and Functions - 2015

The formal feedback I have received is summarised below, where the numbers are average scores given by students on scale of 1-5 with 5 the best. This time there is only one category directly relating to the lecturer: "(The lecturer's) teaching was excellent".

	overall	class size
MATH10101 (Fall 2015)	4.86	229
0C1/1C1 (Fall 2015)	0C1:4.92, 1C1:4.76	105
0C1/1C1 (Fall 2014)	0C1:4.9, 1C1:4.6	109

Teaching relief

I have not yet taken an official sabbatical but I have had various semesters of teaching relief provided by grants etc, including:
• Spring 2026: bought out by Leverhulme Research Project Grant.
• Fall 2025: bought out by Leverhulme Research Project Grant.
• Fall 2022: bought out by RSE Sabbatical Research Grant.
• Spring 2022: bought out by RSE Sabbatical Research Grant.
• Fall 2021: granted relief due to membership of REF2021 panel.
• Spring 2020: bought out by Leverhulme Research Project Grant.
• Spring 2018: bought out by Leverhulme Research Fellowship.
• Fall 2016: bought out by Leverhulme Research Fellowship.

Invited workshops and mini-courses

• Jul 2023: Assouad dimension and fractal geometry - *Mini-course aimed at postgraduate students in fractals*, (delivered online) East China Normal University, Shanghai, China.
• Jul 2017: Fractal geometry and dimension theory - *LMS Undergraduate Summer School*, U. Manchester, UK.
• Aug 2016: How to give a good presentation - *Young Researchers in Mathematics Conference*, U. St Andrews.

PARTICIPATION AND ORGANISATION

Selected meetings and events organised

• 2025: *Harmonic Analysis & PDE network meeting* in St Andrews (with Ana de Orellana).

- 2024: workshop on *Fourier analysis, fractals, and finite fields* in St Andrews.
- 2023: ICMS, EMS, RSE, GMJT funded conference on *Fractal Geometry* at the ICMS in Edinburgh.
- 2019: EMS funded workshop on *Fractal Geometry and Geometric Measure Theory* in St Andrews.
- 2019– : weekly *Analysis Seminar* in St Andrews.
- 2018: Dynamics Workshop during the *British Mathematical Colloquium (BMC)* in St Andrews.
- 2018: LMS funded *one day ergodic theory meeting* in St Andrews (with Mike Todd).
- 2016: *LMS Northern Regional Meeting* and associated two day workshop on *Dynamical systems, ergodic theory and applications* in Manchester.
- 2016: one day *STEM* event in Manchester for Larbert High School, which included the schools of *Mechanical and Civil Engineering*, *Computer Science*, and *Mathematics*.
- 2015–16: weekly *dynamical systems and analysis seminar* in Manchester.
- 2015–16: series of *Pure Mathematics Colloquia* in Manchester (with Gareth Jones and Henrik Suess).
- 2015: LMS funded one day meeting on *Fractal geometry and dimension theory* in Manchester (*celebrating new appointments* grant).
- 2015: LMS funded *one day ergodic theory meeting* in Manchester (with Charles Walkden).

Selected outreach and engagement activities

- Participated in several events or workshops as an invited speaker or facilitator, including:

• Secondary School level: Mathematics in Action (The Training Partnership), U. College London, (audience size: 901) ∘ Mathematics Masterclass, U. York ∘ The Sutton Trust, Uni. St Andrews ∘ First Chances Project, Maths and Science Workshop, U. St Andrews ∘ Making Maths at Manchester, Uni. Manchester ∘ Highland Senior Maths Weekend, Lagganlia (2013–16) ∘ Taking Maths Further, Uni. Manchester ∘ International Science Summer School, Uni. St Andrews ∘ Summer Academic Experience Course, U. St Andrews ∘ Scottish Mathematical Challenge Award Ceremony ∘ visits to give lectures or run workshops at: St Leonard's College, Larbert High School, Glenalmond College, The Bolton School, St Ninian's High School.

• General audience: St Andrews Bright Club ∘ Pint of Science, Edinburgh. ∘ Scottish Mathematical Council conference ∘ public lecture at the ICMS, Edinburgh.

• Undergraduate level: LMS Undergraduate Summer School, Uni. Manchester ∘ Research in the UK afternoon, Uni. Cambridge ∘ St Andrews Student Radio (to discuss life as an academic) ∘ LMS Undergraduate Summer School, The University of Manchester, UK.

• Postgraduate level: Young Researchers in Mathematics Conference, U. Oxford ∘ Lunchtime Legends, Uni. St Andrews ∘ LMS Early Career Researcher Continuing Professional Development panellist: Writing and publication.

SELECTED TALKS

Large international conferences

• The Assouad dimension analogue of the Falconer distance problem: 03/09/24, *Workshop on the Geometry of Deterministic and Random Fractals II*, Rényi Institute, Budapest, Hungary.
• Applications of the Fourier spectrum: 10/06/24, *On the Interface of Geometric Measure Theory and Harmonic Analysis*, Banff, Canada.
• The Fourier spectrum and applications to exceptional sets: 14/05/24, *Madison Lectures in Harmonic Analysis*, University of Wisconsin-Madison, USA.
• Hölder solutions to the winding problem: 12/12/19, *Thermodynamic Formalism: Dynamical Systems, Statistical Properties and their Applications*, Luminy, France.
• Interpolating between dimensions: 01/10/18, Opening Keynote, *Fractal Geometry and Stochastics VI*, Germany.
• Fourier transforms of measures on the Brownian graph: 10/03/16, ICERM workshop *Fractal Geometry, Hyperbolic Dynamics and Thermodynamical Formalism*, Brown University, USA.
• Recent progress on the Assouad dimension: 21/09/15, *Fractals and Related Fields III*, Porquerolles Island, France.
• The Assouad dimension of self-similar sets with overlaps: 09/06/14, *Workshop on Fractals*, Jerusalem, Israel.
• Inhomogeneous iterated function systems: 26/03/14, *Fractal Geometry and Stochastics V*, Tabarz, Germany.
• Modified singular value functions and self-affine carpets: 11/12/12, *Advances in Fractals and Related Topics*, Chinese University of Hong Kong, Hong Kong.
• The Hausdorff dimension of graphs of prevalent continuous functions: 16/06/11, *Fractals and Related Fields II*, Porquerolles Island, France.

Smaller international meetings and workshops

• Averages of the Fourier transform in finite fields: 23/05/25 (online), invited talk, Workshop on Combinatorial and Additive Number Theory, CUNY Graduate Center, NY USA.
• Fourier dimension of the Brownian graph: 31/01/25 (online), Keynote Talk, workshop on Fractals: Geometry, Analysis, Algebra, Numerics and Applications at IIIT Allahabad, India.
• Kleinian group actions and parabolic rational maps: 17/02/24 (online), Plenary Talk, Conference on Functional Analysis and Fractals, February 2024, at IIIT Allahabad, India.
• Assouad dimension of distance sets: 14/05/22 (online), invited speaker, AMS Sectional Meeting - Special Session on Fractal Geometry and Dynamical Systems.
• Assouad dimension of distance sets: 05/12/21 (online), invited speaker, Winter Meeting of the Canadian Mathematical Society.
• A new perspective on the Sullivan dictionary via the Assouad dimension and spectrum: 17/12/20 (online), invited speaker, *Séminaire Cristolien d'Analyse Multifractale*, Laboratoire d'Analyse et de Mathématiques Appliquées, CNRS. Paris, France.
• Box dimensions of projections and dimension interpolation: 17/01/20, invited speaker, *Geometric Measure Theory in Padova*, Italy.
• The Assouad spectrum: 05/12/17, invited speaker, *Fractals and Dimensions*, Mittag-Leffler Institute, Stockholm, Sweden.
• Dimensions of equilibrium measures on some self-affine sets: 11/07/17, invited speaker, *Thermodynamic formalism - Applications to geometry and number theory* , Bremen, Germany.
• Scaling scenery of $(\times m, \times n)$ invariant measures: 23/05/14, invited speaker, *Numbers in Ergodic Theory*, Leiden, The Netherlands.
• On the analyticity of Falconer's subadditive pressure function: 12/05/14, *Conference in honour of Kenneth Falconer's 60th birthday*, INRIA, Paris, France.
• Fourier transforms of measures supported on graphs: 24/01/14, invited speaker, *Ergodic Theory and Dynamical Systems mini-workshop*, Jagiellonian University, Krakow, Poland.

• Random self-affine multifractal Sierpiński sponges in $\mathbb{R}^d$: 14/10/10, *CODY Workshop*, Warsaw, Poland.
• The visibility conjecture: 09/03/10, *Winter School on operators and fractals*, Siegmundsburg, Germany.

Conferences and workshops in the UK

• Subtle horoball asymptotics recording parabolics: 26/03/26 *Statistics, hyperbolicity and related properties (SHARP)*, University of Warwick.
• Incidence geometry over finite fields and exceptional projections: 13/06/25 *Diophantine approximation and related fields*, University of York.
• Fourier analytic techniques in finite fields: 18/09/24, *Fourier analysis, fractals, and finite fields*, University of St Andrews.
• Fourier analytic tools in geometric measure theory and dimension interpolation: 31/04/23, invited speaker, *Paris-London Analysis seminar*, University College London.
• Dimension interpolation in conformal dynamics: 14/09/22, invited speaker, *Geometry, Stochastics, & Dynamics*, Imperial College London.
• An expanded Sullivan dictionary via dimension interpolation: 15/06/22, invited speaker, *One day ergodic theory meeting*, University of Exeter.
• Dimensions of limit sets of Kleinian groups: 06/09/18, invited talk at *Dynamics Days Europe*, Loughborough University.
• Dynamics on fractals (public lecture): 19/06/18, invited public lecture, *Thermodynamic formalism in dynamical systems*, ICMS.
• Ito drift-diffusion processes and applications to Fourier decay problems on random graphs: 31/05/18, invited speaker, *Sixth Scottish Partial Differential Equation Colloquium*, University of Edinburgh.
• Regularity of measures: 11/12/17, invited speaker, *LMS Network on Harmonic Analysis and PDEs*, University of Warwick.
• (Almost) arithmetic progressions and dimension: 29/06/17, invited speaker, *Diophantine approximation and related fields*, University of York.
• Fourier decay of measures on the Brownian graph: 23/10/15, invited speaker, *LMS Harmonic Analysis and PDEs network meeting*, University of Edinburgh.
• Dynamically defined fractals: 17/08/15, invited keynote speaker for dynamics, *Young Researchers in Mathematics Conference*, 17/08/15 - 20/08/15, University of Oxford.
• Fourier analytic properties of the Brownian graph: 22/07/15, *Fractal geometry and dimension theory one day meeting*, University of Manchester.
• The Kakeya Problem: invited plenary speaker for analysis 13/05/15, *Birmingham Young Mathematician Colloquium*, University of Birmingham.
• How to prove Falconer's distance set conjecture using ergodic theory: 03/12/14, invited speaker, *One day ergodic theory meeting*, Queen Mary University of London.
• Dimension and measure for typical random fractals: 16/04/12, *EPSRC Workshop*, University of Warwick.
• The horizon problem for prevalent surfaces: 19/04/11, *EPSRC Workshop*, University of Warwick.

Invited seminars and colloquia abroad

• Dimension interpolation and the geometry of fractals: 03/03/26, Boole Colloquium in Mathematics, University College Cork, Ireland.
• Dimension interpolation and the geometry of fractals: 23/01/26, Distinguished Lecture (online), International Seminar on Quantization, Geometry, and Dynamics.
• Assouad dimension of limit sets of Kleinian groups: 10/04/24, Algebra & Geometry seminar (online), University of New Mexico, Albuquerque, USA.
• Assouad dimension and the Falconer distance problem: 14/07/23, colloquium (online), East China Normal University, Shanghai, China.
• Sullivan's dictionary, conformal dynamics, and Assouad type dimensions: 10/11/22, Fractals seminar (online), Shenzhen Technology University, China.

• A new perspective on the Sullivan dictionary: 3/11/20, Dynamics Seminar (online), Hebrew University of Jerusalem, Israel.
• Dimension theory and the Sullivan dictionary: 29/10/20, Dynamics Seminar (online), Uni. Vienna, Austria
• Parabolic points, Poincaré exponents, and Patterson-Sullivan measures: 30/04/20 (online), Wesleyan University Colloquium, USA.
• Dimensions of Kleinian limit sets: 27/03/19, Stony Brook University, New York, USA.
• The Assouad Spectrum: 23/03/18, University of Waterloo, Canada.
• (Almost) arithmetic progressions, weak tangents, and Assouad dimension: 14/09/17, Chinese University of Hong Kong.
• Fourier transforms of measures on the Brownian graph: 11/09/17, Chinese University of Hong Kong.
• The Assouad spectrum: scaling and homogeneity in metric spaces: 20/10/16, University of Vienna, Austria.
• Dimensions of self-affine carpets and Bernoulli convolutions: 18/07/16, Wesleyan University, USA.
• Inhomogeneous self-similar sets with overlaps: 17/03/16, Ohio State University, USA.
• The Brownian graph has maximal Fourier dimension: 31/08/15, University of Oulu, Finland.
• Fourier transforms of measures on the Brownian graph: 25/08/15, University of Helsinki, Finland.
• Fourier transforms of measures supported on graphs: 12/08/14, Cornell University, USA.
• Generic dimensions of graphs and images: 07/12/12, East China Normal University, Shanghai, China.
• The visible part of fractal sets: 03/03/10, University of Bremen, Germany.

Invited seminars and colloquia in the UK

• Dimension interpolation and the geometry of fractals, 06/02/26, Colloquium, Mathematics Institute, University of Warwick.
• Dimension interpolation, 29/10/25, Colloquium, Department of Mathematical Sciences, Loughborough University.
• Dimensions of orbital sets in conformal dynamics, 27/10/25, dynamics seminar, University of Manchester.
• Orbital sets in conformal dynamics, 16/05/25, dynamics seminar, University of Bristol.
• L^2 restriction estimates from the Fourier spectrum, 12/02/25, Maxwell Institute Analysis Seminar, Edinburgh.
• The Assouad spectrum of Gatzouras-Lalley carpets, 08/01/24, dynamics seminar, University of Bristol.
• Dimensions of parabolic Julia sets and Kleinian limit sets, 07/09/22, dynamics seminar, University of Liverpool.
• Intermediate dimensions of infinitely generated self-conformal sets, 08/02/22 (online), Queen Mary Uni. London
• Dimensions of exceptional self-affine sets in $\mathbb{R}^3$, 12/11/19, analysis seminar, University of Birmingham.
• Kleinian groups, parabolic points, and dimension, 24/10/19, dynamics seminar, University of Bristol.
• Approximate arithmetic structure in large sets of integers: 23/05/19, Imperial College London.
• The Assouad dimension of Kleinian limit sets: 16/05/19, Colloquium, Open University, Milton Keynes.
• Sets of integers containing (almost) arithmetic progressions: 24/10/18, Colloquium, University of Surrey.
• Dimension and measure for limit sets of Kleinian groups: 20/02/18, University of Warwick.
• Regularity of Kleinian limit sets and Patterson-Sullivan measures: 12/02/18, University of Manchester.
• Geometrically finite Kleinian groups and dimension: 09/02/18, University of Liverpool.
• (Almost) arithmetic progressions and dimension: 21/03/17, University of Glasgow.
• (Almost) arithmetic progressions and dimension: 27/02/17, University of Edinburgh.
• Exact dimensionality and Ledrappier-Young formulae for Gibbs measures on some planar self-affine sets: 31/05/16, University of Warwick.
• Brownian motion, Fourier transforms, and dimension: 29/04/16, Pure Maths Colloquium, Uni. Manchester.
• Inhomogeneous self-similar sets with overlaps: 21/04/16, University of Bristol.
• Inhomogeneous attractors: structure and dimension: 06/10/15, Open University, Milton Keynes.
• Dimension theory for parabolic limit sets: 16/06/15, University of Warwick.
• Scaling scenery and the distance set problem: 19/05/15, University of York.
• Equality of Hausdorff measure and Hausdorff content: 06/05/15, University of Liverpool.
• Blowing up ergodic measures: 19/03/15, Imperial College London.
• On the dimensions of a family of overlapping self-affine carpets: 26/02/15, University of Bristol.

• The Assouad dimension of self-similar sets with overlaps: 11/03/14, University of St Andrews.
• On the L^q-spectrum of planar self-affine measures: 19/11/13, University of Manchester.
• Assouad type dimensions and homogeneity of fractals: 01/04/13, University of Edinburgh.
• Assouad type dimensions and homogeneity of fractals: 21/02/13, University of Bristol.

Educational and outreach: University level

• Arithmetic progressions in large sets of integers, 24/02/26, *St Andrews Undergraduate Maths Society (SUMS).*
• Point–line incidences, 23/09/25, *St Andrews Undergraduate Maths Society (SUMS).*
• Conformal dimension: 23/09/25, *Pure maths colloquium pre-talk*, supporting colloquium by Roope Anttila, aimed as PhD students.
• Fractal geometry, dimension, and the space between: 18/02/25, *St Andrews Undergraduate Maths Society (SUMS).*
• Fourier analysis, restriction theory, and $\sqrt{n}$ cancellation: 05/11/24, *Pure maths colloquium pre-talk*, supporting colloquium by Dominique Maldague, aimed as PhD students.
• Prime time mathematics: 26/09/24, *St Andrews Undergraduate Maths Society (SUMS).*
• Fractals: 26/10/23, *How mathematics at St Andrews is transforming our future*, Inner Temple, Crown Office Row, London.
• Arithmetic progressions and an epsilon of room: 26/09/23, *St Andrews Undergraduate Maths Society (SUMS).*
• Engaging with the Public - schools and public engagement: 06/04/22, *Lunchtime Legends*, University of St Andrews.
• Getting Published (Sciences): 27/01/21, *Lunchtime Legends*, University of St Andrews.
• An analyst's take on the integers: 18/11/20, *St Andrews Undergraduate Maths Society (SUMS).*
• Engaging with the Public - schools and public engagement: 22/04/20, *Lunchtime Legends*, University of St Andrews.
• Engaging with the Public - schools and public engagement: 22/04/20, *Lunchtime Legends*, University of St Andrews.
• Fractals and the hidden geometry of nature: 21/05/19, *Pint of Science*, The Canons' Gate, Edinburgh.
• Undergraduate workshop on 'Fractal geometry and dimension theory': July 2017, *LMS Undergraduate Summer School*, University of Manchester.
• An academic career?: 08/03/17, *Careers afternoon*, University of St Andrews.
• How thick is your metric space?: 22/09/16, *St Andrews University Mathematics Society.*
• Workshop on 'How to give a good presentation': 01/08/16, *Young Researchers in Mathematics Conference*, University of St Andrews.
• Patterns in the primes: 11/05/16, *University of Manchester Mathematics Society (Galois Group).*
• The academic career - from PhD to first lectureship: 01/06/15, *Career Development Forum*, University of Manchester.
• Pursuing a career in academia after your PhD: 27/01/15, *Postgraduate Interdisciplinary Mathematics Symposium*, The Burn House.
• Fractal geometry and an exploration of self-similarity: 05/11/14, *University of Manchester Mathematics Society (Galois Group).*
• Fractal geometry and dimension theory: 01/11/12, *Research in the UK afternoon*, Cambridge.
• Furstenberg's proof of the infinitude of the primes: 25/10/12, *St Andrews University Mathematics Society.*
• Fractals: 27/02/09, *St Andrews Undergraduate Maths Society (ChaoSoc).*

Educational and outreach: secondary school level

• Prime numbers, 24/09/25, *Maths Week Scotland*, St Leonard's College.
• Fractals!, 19/06/25, *Scottish Mathematical Challenge Award Ceremony*, St Andrews.
• What is a fractal and how to spot one, 11/07/23, *Summer Academic Experience Course*, St Andrews.
• S3 Experience Day - Subject Speed Dating Session (maths rep.), 18/05/23, St Andrews.
• Fractals in Nature and Science: 06/03/21, *Scottish Mathematical Council Conference*, online.

• Prime numbers and internet security: 12/02/20, *First Chances Project, Maths and Science Workshop*, University of St Andrews.
• Fractals and problem solving: 13/02/19, *First Chances Project, Maths and Science Workshop*, University of St Andrews.
• Fractals: 14/09/18, St Leonard's College.
• What does a broccoli have in common with the stock market?: 05/07/18, *Sutton Trust*, St Andrews.
• Arithmetic of prime numbers, 07/06/18, *Scottish Mathematical Challenge Award Day*, St Andrews.
• Fractals: 17/03/18, *Mathematics Masterclass*, University of York.
• Fractals and problem solving: 14/02/18, *First Chances Project, Maths and Science Workshop*, University of St Andrews.
• Mathematics, fractals and gnome related problem solving: 16/03/17, *STEM week*, Larbert High School.
• Fractals: patterns in nature and science: 26/01/17, *Academic Enrichment Programme*, Glenalmond College.
• Mathematics, fractals and problem solving: 18/11/16, *Highland Senior Maths Weekend*, Lagganlia.
• Why study mathematics?: 27/06/16, *Taking Maths Further*, University of Manchester.
• Fun with Fractals: 03-04/02/16, *University of Manchester Mathematics Interview Day.*
• Fractals and exploring the infinite: 01/02/16, The Bolton School.
• Fractals: describing nature with mathematics: 03/12/15, invited speaker, *Mathematics in Action (The Training Partnership)*, University College London, (audience size: 901).
• Fractals, problem solving, and fun with primes: 13/11/15, *Highland Senior Maths Weekend*, Lagganlia.
• What is mathematics all about?: 25/06/15, *Taking Maths Further*, University of Manchester.
• Mathematics beyond calculation: 25/03/15, *University of Manchester Mathematics Open Day.*
• Fractals, problem solving, and mathematical proof: 08/11/14, *Highland Senior Maths Weekend*, Lagganlia.
• What does a mathematician do?: 09/10/14, *Gymnasium Eppendorf (Hamburg, Germany) field trip*, Manchester.
• A brief introduction to fractals: 09/11/13, *Highland Senior Maths Weekend*, Lagganlia.
• Fractal geometry and problem solving in mathematics: 19/07/13, *International Science Summer School*, St Andrews.
• Fractals and other fun mathematics: 03/07/13, *Sutton Trust*, St Andrews.

PUBLICATIONS

Books

(1) Assouad Dimension and Fractal Geometry, *CUP Tracts in Mathematics Series*, **222**, (2020).

Survey and expository papers

(9) An invitation to dimension interpolation, arXiv:2603.10729
(8) L^p averages of the discrete Fourier transform and applications (with F. Rakhmonov), arXiv,
Combinatorial and Additive Number Theory VII, Springer Proceedings in Mathematics & Statistics (to appear)
(7) Applications of dimension interpolation to orthogonal projections,
Research in the Mathematical Sciences, **12**, (2025), 23 pp.
(6) Inhomogeneous attractors and box dimension,
Contemporary Mathematics, **825**, (2025), 67–85.
(5) The Poincaré exponent and the dimensions of Kleinian limit sets,
The American Mathematical Monthly, **129**, (2022), 480–484.
(4) Fractal geometry of Bedford-McMullen carpets,
Proceedings of the Fall 2019 Jean-Morlet Chair programme, Springer Lecture Notes Series, 2021
(Eds. M. Pollicott & S. Vaienti).
(3) Interpolating between dimensions,
Fractal Geometry and Stochastics VI, Birkhäuser, Progress in Probability, 2021,
(Eds. U. Freiberg, B. Hambly, M. Hinz & S. Winter).
(2) Almost arithmetic progressions in the primes and other large sets,
The American Mathematical Monthly, **126**, (2019), 553–558.
(1) Sixty Years of Fractal Projections, (with K. J. Falconer & X. Jin)
Fractal Geometry and Stochastics V, Birkhäuser, Progress in Probability, 2015,
(Eds. C. Bandt, K. J. Falconer & M. Zähle).

Research papers - under review

(92) On Fourier decay and the distance set problem, 29 pages (with T. Pham), arXiv:2604.19486
(91) Badly approximable points on non-linear carpets, 23 pages (with R. Anttila and H. Koivusalo), arXiv:2603.11822
(90) Fourier restriction for the additive Brownian sheet, 18 pages (with A. E. de Orellana), arXiv:2603.11822
(89) The Assouad spectrum and dimension of typical graphs, 18 pages (with T. Feng), arXiv:2603.11822
(88) Dimensions of orbital sets in complex dynamics (with Y. Xu), 21 pages arXiv:2603.11822
(87) Hausdorff dimension of restricted Kakeya sets, 21 pages arXiv:2505.05709 (with L. Yang),
(86) L^2 restriction estimates from the Fourier spectrum, 28 pages arXiv:2412.14896 (with M. Carnovale and A. E. de Orellana),

Research papers - accepted

(85) An improved L^2 restriction theorem in finite fields, 12 pages arXiv:2505.09293 (with F. Rakhmonov),
Proceedings of the American Mathematical Society
(84) L^p averages of the Fourier transform in finite fields, 31 pages, arXiv:2407.08589
Indiana University Mathematics Journal
(83) Fourier decay of product measures, 17 pages, arXiv:2405.05878
Illinois Journal of Mathematics
(82) A Fourier analytic approach to exceptional set estimates for orthogonal projections, 18 pages, arXiv:2404.11179, (with A. E. de Orellana)
Indiana University Mathematics Journal

Research papers - published

2026

(81) Sobolev and quasiconformal distortion of intermediate dimension with applications to conformal dimension, (with J. T. Tyson),
Journal of the London Mathematical Society, **113** (2026), Paper No. e70445, 58 pp.

(80) Fourier analytic properties of Kakeya sets in finite fields,
Bulletin of the London Mathematical Society, **58** (2026), Paper No. e70367.

(79) Exceptional projections in finite fields: Fourier analytic bounds and incidence geometry, (with F. Rakhmonov),
Mathematische Zeitschrift, **312** (2026), Paper No. 111.

(78) Obtaining the Fourier spectrum via Fourier coefficients, (with M. Carnovale and A. E. de Orellana)
Proceedings of the American Mathematical Society, **154** (2026), 2005–2017.

(77) Assouad type dimensions of parabolic Julia sets, (with L. Stuart)
Mathematische Zeitschrift, **312** (2026), Paper No. 108.

(76) On variants of the Furstenberg set problem,
Proceedings of the American Mathematical Society, **154** (2026), 1093–1103.

(75) Assouad spectrum of Gatzouras-Lalley carpets, (with A. Banaji, I. Kolossváry and A. Rutar)
Advances in Mathematics, **484** (2026), Paper No. 110707, 46 pp.

2025

(74) Critical exponents and dimension for generalised limit sets, (with T. Feng)
Nonlinearity, **38**, (2025), 125018.

(73) Parabolic carpets, (with N. Jurga)
Israel Journal of Mathematics, **267**, (2025), 583-621.

2024

(72) The Fourier spectrum and sumset type problems,
Mathematische Annalen, **390**, (2024), 3891-3930.

(71) A new perspective on the Sullivan dictionary via Assouad type dimensions and spectra, (with L. Stuart)
Bulletin of the American Mathematical Society, **61**, (2024), 103–118.

(70) Assouad type dimensions of infinitely generated self-conformal sets, (with A. Banaji)
Nonlinearity, **37**, (2024), Paper No. 045004.

(69) Assouad-type dimensions of overlapping self-affine sets, (with A. Rutar)
Annales Fennici Mathematici, **49**, (2024), 3–21.

2023

(68) A nonlinear projection theorem for Assouad dimension and applications,
Journal of the London Mathematical Society, **107**, (2023), 777–797.

(67) Intermediate dimensions of infinitely generated attractors, (with A. Banaji)
Transactions of the American Mathematical Society, **376**, (2023), 2449–2479.

(66) Box dimensions of $(\times m, \times n)$-invariant sets, (with N. Jurga)
Indiana University Mathematics Journal, **72**, (2023), 2341–2367.

(65) The Assouad dimension of self-affine measures on sponges, (with I. Kolossváry)
Ergodic Theory and Dynamical Systems, **43**, (2023), 2974–2996.

(64) Diophantine approximation in metric space, (with H. Koivusalo and F. Ramírez)
Bulletin of the London Mathematical Society, **55**, (2023), 756–776.

(63) Refined horoball counting and conformal measure for Kleinian group actions, (with L. Stuart)
Annales Fennici Mathematici, **48**, (2023), 325–344.

(62) The Assouad spectrum of Kleinian limit sets and Patterson-Sullivan measure, (with L. Stuart)
Geometriae Dedicata, **217**, (2023), Paper No. 1, 32 pp.

(61) Dimensions of Kleinian orbital sets, (with T. Bartlett)
Journal of Fractal Geometry, **10**, (2023), 267–278.

(60) Minkowski dimension for measures, (with K. J. Falconer & A. Käenmäki)
Proceedings of the American Mathematical Society, **151**, (2023), 779–794.

2022

(59) On the Fourier dimension of (d, k)-sets and Kakeya sets with restricted directions, (with T. Harris & N. Kroon)
Mathematische Zeitschrift, **301**, (2022), 2497–2508.

(58) Dimensions of the popcorn graph, (with H. Chen & H. Yu)
Proceedings of the American Mathematical Society, **150**, (2022), 4729–4742.

(57) The fractal structure of elliptical polynomial spirals, (with S. A. Burrell & K. J. Falconer)
Monatshefte für Mathematik, **199**, (2022), 1–22.

2021

(56) The box dimensions of exceptional self-affine sets in $\mathbb{R}^3$, (with N. Jurga)
Advances in Mathematics, **385**, (2021), 107734.

(55) L^q-spectra of measures on planar non-conformal attractors, (with K. J. Falconer & L. D. Lee)
Ergodic Theory and Dynamical Systems, **41**, (2021), 3288–3306.

(54) On Hölder maps and prime gaps, (with H. Chen)
Real Analysis Exchange, **46**, (2021), 523–532.

(53) The Assouad spectrum of random self-affine carpets, (with S. Troscheit)
Ergodic Theory and Dynamical Systems, **41**, (2021), 2927–2945

(52) Approximate arithmetic structure in large sets of integers, (with H. Yu)
Real Analysis Exchange, **46**, (2021), 163–173.

(51) Assouad dimension influences the box and packing dimensions of orthogonal projections, (with K. J. Falconer & P. Shmerkin)
Journal of Fractal Geometry, **8**, (2021), 247–259.

(50) L^q-spectra of self-affine measures: closed forms, counterexamples, and split binomial sums, (with L. D. Lee, I. D. Morris & H. Yu)
Nonlinearity, **34**, (2021), 6331–6357.

(49) Regularity versus smoothness of measures, (with S. Troscheit)
Pacific Journal of Mathematics, **311**, (2021), 257–275.

(48) On Hölder solutions to the spiral winding problem,
Nonlinearity, **34**, (2021), 3251–3270.

(47) Projection theorems for intermediate dimensions, (with S. A. Burrell & K. J. Falconer)
Journal of Fractal Geometry, **8**, (2021), 95–116.

(46) Improved bounds on the dimensions of sets that avoid approximate arithmetic progressions, (with P. Shmerkin & A. Yavicoli)
Journal of Fourier Analysis and Applications, **27**:4, (2021).

(45) Schmidt's game on Hausdorff metric and function spaces: generic dimension of sets and images, (with Á. Farkas, E. Nesharim & D. Simmons)
Mathematika, **67**, (2021), 196–213.

2020

(44) Intermediate dimensions, (with K. J. Falconer & T. Kempton)
Mathematische Zeitschrift, **296**, (2020), 813–830.

(43) Attainable values for the Assouad dimension of projections, (with A. Käenmäki)
Proceedings of the American Mathematical Society, **148**, (2020), 3393–3405.

(42) Dimensions of equilibrium measures on a class of planar self-affine sets, (with T. Jordan & N. Jurga).
Journal of Fractal Geometry, **7**, (2020), 87–111.

(41) On the upper regularity dimensions of measures, (with D. C. Howroyd).
Indiana University Mathematics Journal, **69**, (2020), 685–712.

(40) The dimensions of inhomogeneous self-affine sets, (with S. A. Burrell)
Annales Academiæ Scientiarum Fennicæ Mathematica, **45**, (2020), 313–324.

2019

(39) Regularity of Kleinian limit sets and Patterson-Sullivan measures,

Transactions of the American Mathematical Society, **372**, (2019), 4977–5009.

(38) Dimension growth for iterated sumsets, (with D. C. Howroyd & H. Yu)
Mathematische Zeitschrift, (2019), **293**, 1015–1042.

(37) On the Hausdorff dimension of microsets, (with D. C. Howroyd, A. Käenmäki & H. Yu)
Proceedings of the American Mathematical Society, **147**, (2019), 4921–4936.

(36) Dimensions of sets which uniformly avoid arithmetic progressions, (with K. Saito & H. Yu)
International Mathematics Research Notices, (2019), 4419–4430.

(35) The Assouad spectrum and the quasi-Assouad dimension: a tale of two spectra,
(with K. E. Hare, K. G. Hare, S. Troscheit & H. Yu)
Annales Academiæ Scientiarum Fennicæ Mathematica, **44**, (2019), 379–387.

(34) Inhomogeneous self-similar sets with overlaps, (with S. Baker & Á. Máthé).
Ergodic Theory and Dynamical Systems, **39**, (2019), 1–18.

2018

(33) Distance sets, orthogonal projections, and passing to weak tangents,
Israel Journal of Mathematics, **226**, (2018), 851–875.

(32) Assouad type spectra for some fractal families, (with H. Yu).
Indiana University Mathematics Journal, **67**, (2018), 2005–2043.

(31) New dimension spectra: finer information on scaling and homogeneity, (with H. Yu).
Advances in Mathematics, **329**, (2018), 273–328.

(30) Arithmetic patches, weak tangents, and dimension, (with H. Yu).
Bulletin of the London Mathematical Society, **50**, (2018), 85–95.

(29) The Assouad dimension of randomly generated fractals, (with J. J. Miao & S. Troscheit).
Ergodic Theory and Dynamical Systems, **38**, (2018), 982–1011.

(28) Quantifying inhomogeneity in fractal sets, (with M. Todd).
Nonlinearity, **31**, (2018), 1313–1330.

(27) On the L^q dimensions of measures on Hueter-Lalley type self-affine sets, (with T. Kempton).
Proceedings of the American Mathematical Society, **146**, (2018), 161–173.

(26) On the Fourier analytic structure of the Brownian graph, (with T. Sahlsten)
Analysis & PDE, **11**, (2018), 115–132.

2017

(25) The Assouad dimensions of projections of planar sets, (with T. Orponen)
Proceedings of the London Mathematical Society, **114**, (2017), 374–398.

(24) The Assouad dimension of self-affine carpets with no grid structure, (with T. Jordan)
Proceedings of the American Mathematical Society, **145**, (2017), 4905–4918.

(23) Some results in support of the Kakeya conjecture, (with E. J. Olson & J. C. Robinson)
Real Analysis Exchange, **42**, (2017), 253–268.

(22) Assouad type dimensions for self-affine sponges, (with D. C. Howroyd)
Annales Academiæ Scientiarum Fennicæ Mathematica, **42**, (2017), 149–174.

(21) Uniform scaling limits for ergodic measures, (with M. Pollicott)
Journal of Fractal Geometry, **4**, (2017), 1–19.

2016

(20) On the L^q-spectrum of planar self-affine measures,
Transactions of the American Mathematical Society, **368**, (2016), 5579–5620.

(19) Inhomogeneous self-affine carpets,
Indiana University Mathematics Journal, **65**, (2016), 1547–1566.

(18) On the dimensions of a family of overlapping self-affine carpets, (with P. Shmerkin)
Ergodic Theory and Dynamical Systems, **36**, (2016), 2463–2481.

2015

(17) On the Assouad dimension of self-similar sets with overlaps,
(with A. M. Henderson, E. J. Olson & J. C. Robinson)
Advances in Mathematics, **273**, (2015), 188–214.

(16) Micromeasure distributions and applications for conformally generated fractals, (with M. Pollicott)
Mathematical Proceedings of the Cambridge Philosophical Society, **159**, (2015), 547–566.

(15) First and second moments for self-similar couplings and Wasserstein distances,
Mathematische Nachrichten, **288**, (2015), 2028–2041.

(14) On the equality of Hausdorff measure and Hausdorff content, (with Á. Farkas)
Journal of Fractal Geometry, **2**, (2015), 403–429.

(13) Scaling scenery of $(\times m, \times n)$ invariant measures (with A. Ferguson & T. Sahlsten)
Advances in Mathematics, **268**, (2015), 564–602.

(12) Remarks on the analyticity of subadditive pressure for products of triangular matrices
Monatshefte für Mathematik, **177**, (2015), 53–65.

(11) Dimension and measure for typical random fractals
Ergodic Theory and Dynamical Systems, **35**, (2015), 854–882.

(10) A note on the 1-prevalence of continuous images with full Hausdorff dimension, (with J. T. Hyde)
Journal of Mathematical Analysis and Applications, **421**, (2015), 1713–1720.

2014

(9) On Fourier analytic properties of graphs (with T. Orponen & T. Sahlsten)
International Mathematics Research Notices, (2014), 2730–2745.

(8) Assouad type dimensions and homogeneity of fractals
Transactions of the American Mathematical Society, **366**, (2014), 6687–6733.

2013

(7) Dimension and measure for generic continuous images (with R. Balka, Á. Farkas & J. T. Hyde)
Annales Academiæ Scientiarum Fennicæ Mathematica, **38**, (2013), 389–404.

(6) The visible part of plane self-similar set (with K. J. Falconer)
Proceedings of the American Mathematical Society, **141**, (2013), 269–278.

2012

(5) Inhomogeneous self-similar sets and box dimensions
Studia Mathematica, **213**, (2012), 133–156.

(4) On the packing dimension of box-like self-affine sets in the plane
Nonlinearity, **25**, (2012), 2075–2092.

(3) The Hausdorff dimension of graphs of prevalent continuous functions (with J. T. Hyde)
Real Analysis Exchange, **37**, (2011/2012), 333–352.

2011

(2) Multifractal spectra of random self-affine multifractal Sierpiński sponges in $\mathbb{R}^d$ (with L. Olsen)
Indiana University Mathematics Journal, **60**, (2011), 937–984.

(1) The horizon problem for prevalent surfaces (with K. J. Falconer)
Mathematical Proceedings of the Cambridge Philosophical Society, **151**, (2011), 355–372.

Other publications

(5) Mathematics News Flash,
2021– : Bimonthly article in *Newsletter of the London Mathematical Society*

(4) Fractals all around us,
The Scottish Mathematical Council Journal, **51**, (2020), 53–56.

(3) 2520 ants for Dylan,
Newsletter of the London Mathematical Society, **488**, (2020), 28–30.

(2) Logical puzzles with gnomes and hats,
The Scottish Mathematical Council Journal, **45**, (2015), 48–51.

(1) Fractals in Lagganlia,
The Scottish Mathematical Council Journal, **44**, (2014), 58–60.